

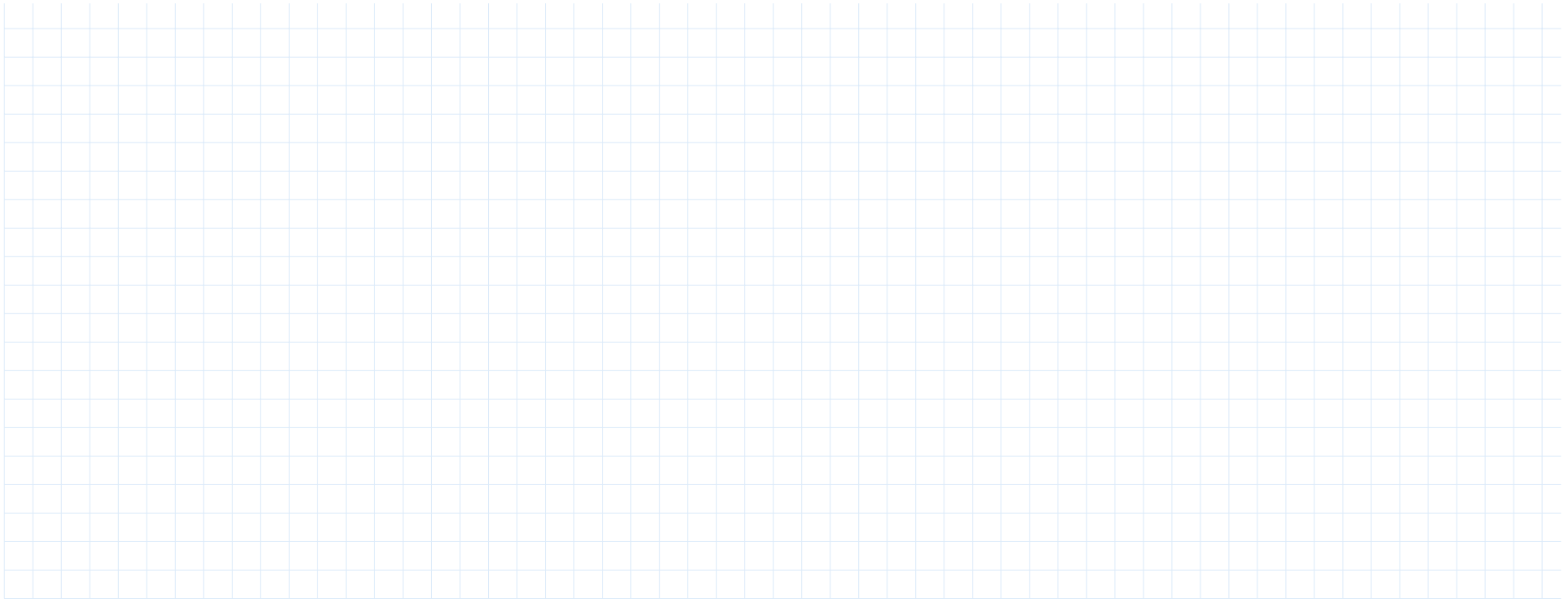
Composition: One Function Inside Another

Sections 4.1-4.2

Composition: $(f \circ g)(x) = f(g(x))$. Apply the inside function first.

$$f(x) = x^2 + 1, \quad g(x) = 2x - 3.$$

Find $(f \circ g)(x)$, $(g \circ f)(x)$, and $(f \circ g)(2)$. Are the two compositions equal?



Composition Domains: Keep Inner Restrictions

Sections 4.1-4.2

An input must be in the domain of the **inside** function, and its output must be allowed in the **outside** function.

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{3x}{x^2 - 1}.$$

Find $(f \circ g)(x)$ and its domain. Do not lose restrictions when simplifying.



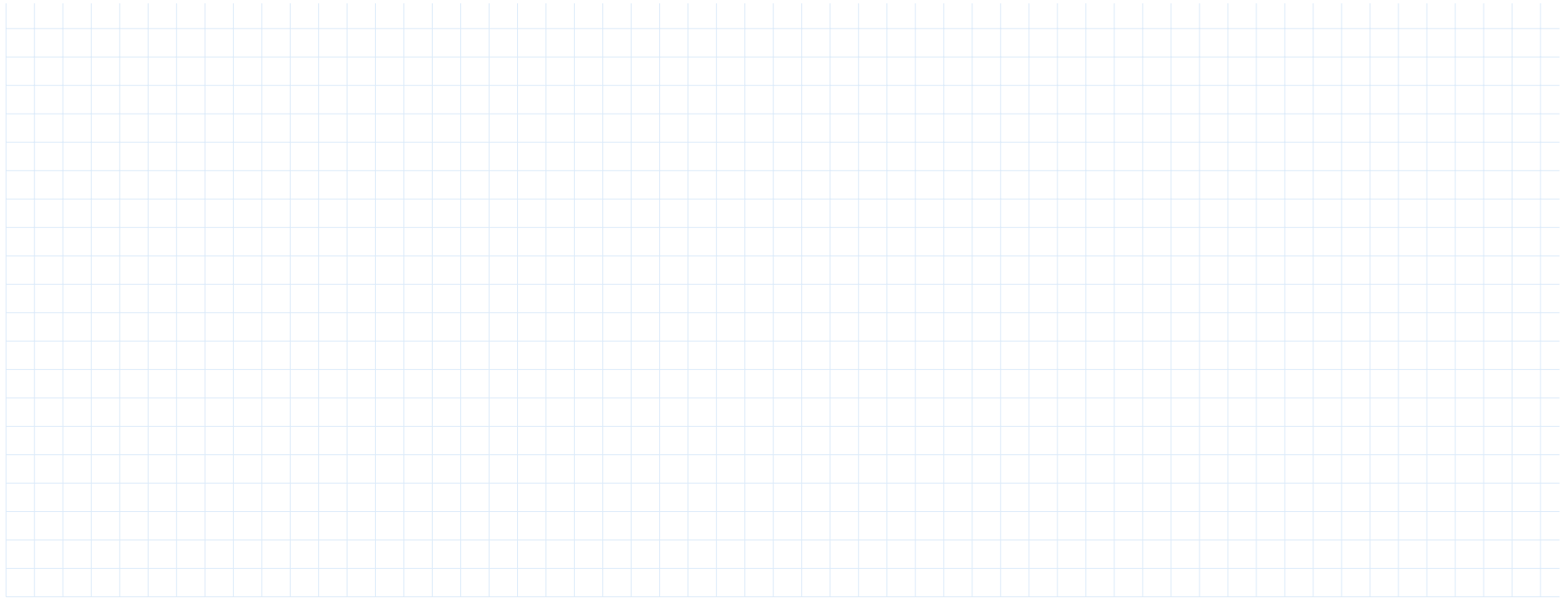
Composition with Square Roots

Let $f(x) = \sqrt{x-2}$ and $g(x) = x^2 - 1$. Find both compositions and the domain of each.

$$(f \circ g)(x)$$

$$(g \circ f)(x).$$

A real square root requires a nonnegative radicand.



One-to-One Functions and Reversed Pairs

Sections 4.1-4.2

A function is **one-to-one** if different inputs have different outputs. Its graph passes the horizontal line test. Decide whether $f(x) = x^2$ and $g(x) = x^3$ are one-to-one on $\mathbb{R}$. Then reverse the pairs below to find F^{-1} , and state the domain and range of each:

$$F = \{(-2, 2), (-1, 1), (0, 0), (1, 3)\}.$$



Find and Check an Inverse

Sections 4.1-4.2

Find f^{-1} for $f(x) = 2x + 5$, then check both compositions. For a second example, find g^{-1} for $g(x) = (x - 2)^3$. **Method:** write $y = f(x)$, interchange x and y , then solve for y .



Restrict a Domain Before Inverting

The full quadratic $x^2 + 2$ is not one-to-one. Use the restricted function

$$f(x) = x^2 + 2, \quad x \geq 0.$$

Find its inverse. State both domains and ranges, and check both compositions on their correct domains.



Inverse Graphs Reflect Across a Line

Sections 4.1-4.2

Sketch $f(x) = x^2 + 2$ for $x \geq 0$, its inverse $f^{-1}(x) = \sqrt{x - 2}$, and the line $y = x$. Use the points $(0, 2)$, $(1, 3)$, and $(2, 6)$ on f to locate three points on its inverse.

